GAMMs: Additive Mixed Models

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Introduction

Generalised additive mixed models (GAMMs) bring together two powerful regression frameworks: smooth non-linear effects from generalised additive models (GAMs) and random effects from linear / generalised linear mixed models. The `mgcv::gam()` interface handles both in a single unified framework by treating random effects as a specific type of penalised smooth, making the combined model no harder to fit than a plain GAM. The result is the natural choice when clustered data also exhibit non-linear effects of continuous covariates — longitudinal trajectories, growth curves, ecological time series with site-level structure.

Prerequisites

A working understanding of generalised additive models (smooth effects, basis functions, penalised regression splines), linear mixed models, and the role of random effects in clustered data.

Theory

In `mgcv`, random effects are smooths with the basis specification `bs = "re"`. A smooth of a grouping factor with this basis is mathematically equivalent to a random intercept; the smoothing penalty plays the role of the random-effect variance, and REML estimation of the smoothing parameter recovers the variance estimate. More complex random structures — random slopes, factor-by-smooth interactions — are constructed via tensor-product smooths or `s(group, bs = "re")` combined with appropriate factor coding.

For non-Gaussian outcomes, change the `family` argument; the random-effect basis trick generalises to GLMM territory automatically.

Assumptions

Smooth-function assumptions (the true relationship is reasonably represented by penalised splines), random-effect Normality, conditional independence given random effects, and the appropriate family for the outcome.

R Implementation

```r
library(mgcv)

set.seed(2026)
d <- data.frame(
  subject = factor(rep(1:20, each = 30)),
  time    = rep(seq(0, 10, length.out = 30), 20)
)
d$y <- rnorm(nrow(d)) + sin(d$time) * 2 +
       rep(rnorm(20, 0, 1.5), each = 30) +
       0.3 * d$time

fit <- gam(y ~ s(time) + s(subject, bs = "re"), data = d)
summary(fit)
plot(fit, pages = 1)
```

Output & Results

`summary(fit)` reports the effective degrees of freedom (edf) for each smooth, parametric coefficients, and an approximate F-test for each smooth term. The random-effect smooth contributes a single variance component summarised in the model summary; `plot(fit, pages = 1)` displays each smooth on a single page.

Interpretation

A reporting sentence: "A GAMM with a smooth of time (edf 7.2, $F = 24.5$, $p < 0.001$) and a subject-level random intercept (variance 2.18) explained 72 % of the deviance; the smooth recovered the simulated sine-plus-linear trend faithfully and the random-intercept SD matched the simulation parameter to within Monte Carlo error." Always report edf for smooths and the random-effect variance.

Practical Tips

- `bs = "re"` turns a smooth into a random effect; `s(group, bs = "re")` is a random intercept; `s(group, x, bs = "re")` is a random slope on x.
- Tensor product smooths (`te(x, z)`) model smooth interactions between two continuous predictors; `ti(x, z)` builds the pure interaction without the main effects, useful for ANOVA-like decomposition.
- For GLMM-type responses (binomial, Poisson, beta), specify `family = binomial(link = "logit")` (etc.) and the same random-effect smooth syntax works.
- Use `mgcv::gam.check(fit)` for diagnostic plots of residuals and basis-dimension adequacy; raise `k` when basis dimensions are too small.
- `gratia::draw(fit)` produces ggplot-style visualisations of all model components, far cleaner than base `plot.gam()` for publications.
- For very large datasets ($n > 10{,}000$), use `mgcv::bam()` with `discrete = TRUE` for substantial speed-ups; the API is otherwise identical.

R Packages Used

`mgcv` for `gam()`, `bam()`, smooth bases, and REML smoothing-parameter selection; `gratia` for ggplot-style GAM diagnostics and component plots; `lme4` for comparative LMM fits; `gamm4` for GAMMs implemented via `lme4` for very large random-effect structures.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures $\rightarrow$ mixed models
- GAMs with mgcv
- Non-linear regression with nls

- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI